AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M218
Relational Databases

Mid-term Examination MTA - Makeup
Fall Semester 2025/2026

Date: 4 /Dec. /2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 1.5 Hours

Instructions:

- Total Marks: 30
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part 1: Multiple choice questions [5 Marks]

This part consists of 5 questions carrying a weight of 1 mark for each.

1. Which level of the three-schema architecture describes the entire structure of the database for a community of users?
   a. External level
   b. Conceptual level
   c. Internal level
   d. View level
2. Which of the following is NOT a property of a relation in the relational model?
   a. Tuples have no inherent ordering
   b. Each attribute value must be atomic
   c. Multivalued attributes are allowed in a single column
   d. Each tuple must be unique
3. Which relational algebra operation extracts specific columns (attributes) from a relation?
   a. SELECTION (σ)
   b. PROJECTION (π)
   c. UNION (∪)
   d. JOIN (⋈)
4. During the database development lifecycle, which stage involves transforming the conceptual model into a logical (relational) model with primary keys and foreign keys defined?
   a. Requirements analysis
   b. Conceptual design
   c. Logical design
   d. Physical design
5. A student's course enrollment data contains repeated groups of attributes in the same row. Which normal form is violated and which process would you apply first?
   a. 1NF violated – Apply decomposition to eliminate multivalued attributes
   b. 2NF violated – Remove partial dependencies
   c. 3NF violated – Remove transitive dependencies
   d. BCNF violated – Split into two relations with candidate keys

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Part 2: Problem solving questions [25 Marks]

Question 1 [8 Marks]

1- Consider the following ER model fragment:

- A Researcher is always assigned to a Project.
- A Researcher can also supervise multiple Projects.
- An inclusion constraint (I) exists between SUPERVISE and ASSIGNED_TO.

Explain what the inclusion constraint implies. Write the relational schemas and indicate the foreign key(s) needed. **[2 marks]**

2- In an ER model, an entity User may be either a Buyer or a Seller, but not both. This is a case of Partial Disjoint Specialization.

Define the type of constraint this represents. Show two possible ways to model this in the relational schema. **[2 marks]**

3- An entity Transaction must be related to either a Product Sale or a Service Request, but never both. There is an exclusive arc connecting them.

Identify the type of constraint being modeled. Translate this situation into relational tables. **[2 marks]**

4- In a university database, every Assignment Submission must be linked to either an Individual Student or a Group, but not both. A disjoint total specialization exists in this case.

What does the disjoint total specialization mean here? Show how you would implement this specialization in a relational schema. Explain what the constraint ensures. **[2 marks]**

Question 2 [8 Marks]

Consider the following relational schema for a library system. For each relation, the attributes in **bold** constitute the key:

Member(**MemberID**, MemberName, Age, Gender)

Book(**BookID**, Title, Author, Genre)

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Loan(**MemberID, BookID**, LoanDate, ReturnDate)

Librarian(**LibrarianID**, LibrarianName, Department)

Write the following queries in **relational algebra**:

1- List the titles of books borrowed by female members whose age is above 25. **[2 marks]**
2- Retrieve the names of members who borrowed at least two different books by "J.K. Rowling". **[2 marks]**
3- Find the titles of books that were never borrowed by any member above age 30. **[2 marks]**
4- Get the names of members who borrowed at least one book from each of the genres: "History" and "Philosophy". **[2 marks]**

Question 3 [9 Marks]

- A training center offers various certified workshops led by instructors and attended by registered participants.
- Each Instructor has a unique InstructorID, Name, and Expertise.
- Each Workshop has a unique WorkshopID, Title, Duration (in hours), and Level (e.g., Beginner, Intermediate, Advanced).
- Each Participant has a unique ParticipantID, FullName, Email, and DateOfBirth.
- A Participant can attend multiple Workshops, and each Workshop can have multiple Participants.
- Each Participation is recorded with ParticipationID, EnrollmentDate, and a final Score.
- Each Workshop is conducted by exactly one Instructor, but each Instructor can conduct many Workshops.

1- Identify the entities, their attributes, and primary keys. (List all main entities and their attributes. Clearly identify the PKs.) **[3 marks]**
2- Draw the ER diagram representing the above scenario. (Include relationships and specify cardinalities. Indicate PKs and FKs.) **[3 marks]**
3- Convert the ER diagram into relational schemas (table structures). (Show each relation with primary and foreign keys clearly marked.)**[3 marks]**

End of questions

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